TEAM SEVEN ATE NINE

Arm Slot machine II presentation

Research

- Slot machine design language, animations, layout
  - Multiple reels, Colorful layout, Multiple slot machines
- Features commonly seen in slot machines
  - Login system, Persistent balance across devices
- General good frontend coding practices
  - Responsive, Good animations, Consistent formatting
- General good backend coding practices
  - Separation of concerns, Meaningful variable and function names
- Codex how to properly use
  - AGENTS.md, SKILLS.md, MCP Configuration
- How to keep users happy
  - Sound effects, Wins should be rewarding, Losses should be quick

Initial Strategy

- We equipped Codex with the Playwright skill to allow for it to check what it's doing with a browser.
- We also made AGENTS.md with all general requirements and guidelines for Codex
- We also equipped Codex with skills to create PRs and resolve issues we make on github
- We will be using GPT-5.4 with medium reasoning.

We will be focusing on giving it very well specified issues on github so that it can execute and open a PR on them.

Primary Objective

- A fully fledged Slot Machine website with a pirate theme
- It will have a variety of features commonly seen in other slot machines + features to keep player engagement
  - Cache login with password encryption + guest login
  - Soundtrack and sound effects
  - Animations
  - Leaderboard system
  - Buy cosmetics with tokens won.

Tech Stack

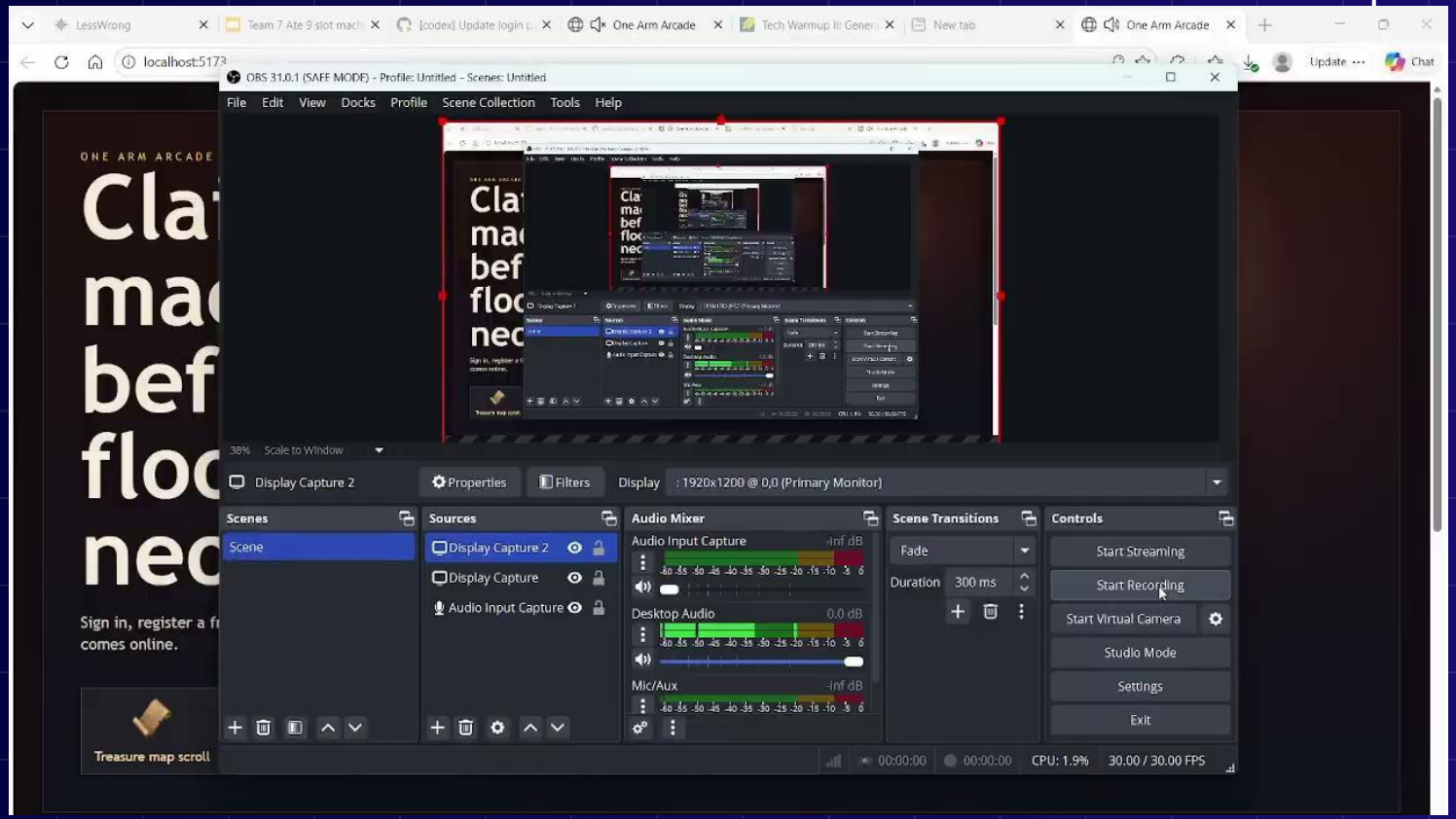
We used the MERN tech stack for the following reasons

- MongoDB - Lightweight database to store login information, account balances, leaderboard data
- Express - Simplest serving middleware our team knew
- React - Modular and easy frontend library that our team could easily understand
- Node.JS - The classic backend system that allows all the other 3 technologies to easily integrate

Implementation

- We created issues on github for independent features
- Then we had codex read the issues to create pull requests
- Imported pirate themed assets and had the codex integrate it into the website

Problems we faced

- Codex produced UI which was not good looking
- On complex tasks, codex usually ignored the general guidelines and sometimes forgot to push changes and create a PR
- Utilized front-end skill mainly, still made a not great looking frontend
- When given many instructions in a github issue, it only chose to follow a few
- Did not give an explanation why it did not do the issue items!

This is the Final Product!